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RESEARCH ARTICLE

Optimization of Fault Identification and Location Using Adaptive Neuro-Fuzzy Inference System and Support Vector Machine for an AC Microgrid

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ABSTRACT Conventional methods for high-impedance faults, low fault current levels, and communication delays could not properly identify the fault identification and location of an AC Microgrid. Fault identification and locating are crucial when integrating renewable energy sources with AC Microgrids. In an AC Microgrid, high-impedance faults, low-fault current levels, and communication delays are incapable approaches for fault identification and fault location. Machine learning enables rapid fault identification and location. This paper develops an adaptive neuro-fuzzy inference system and a support vector machine approach. To address these issues, lower fault current levels, detect high-impedance faults and affect communication delays. The proposed method is tested and evaluated in IEEE12BUS system, both in islanded and grid-connected modes, with execution times of 0.00202s in islanded mode and 0.0022s in grid-connected mode. The proposed adaptive neuro-fuzzy inference system method recognizes the optimal fault type. At the same time, Support vector machine identifies fault location accurately, resulting in the shortest execution time and minimal error percentage. This approach is demonstrated by using the IEEE12BUS AC Microgrid system. Hence, this approach is well-suited for real-time applications in AC Microgrid systems.

INDEX TERMS Alternate current (AC), microgrid (MG), renewable energy source (RES), support vector machine (SVM), adaptive neuro-fuzzy inference system (ANFIS), machine learning (ML), fault identification (FI), faulty location (FL).

I. INTRODUCTION

In electrical networks, renewable energy sources are increasingly the main suppliers of power. This is because of the rising demand for energy, problems with electricity quality, and the damaging impact of fossil fuels on the environment. Furthermore, it alters the traditional network's faulty current levels and direction [1]. Because they facilitate the integration of distributed generation (DG) in distribution networks (DNs) with strict standards for power quality and efficiency, microgrids are important in this scenario.

Microgrids can make it easier to carry out several smart grid features, including distributed generation, self-monitoring, adaptive and islanding mode, digital and two-way communication, and remote checks. The benefits of MG's are utilized in a wide range of applications, including cyber security, energy management, control, monitoring, and most importantly, faulty identification and location to ensure a dependable and secure estimation.

The AC microgrid is easily compatible with the main grid and the existing conventional house appliances, which are the focus of this research work. distributed generation (DG) and load-containing electricity distribution systems, or renewable microgrids (MGs), can facilitate popular access to DG and renewable energy [2].

However, implementing microgrids into practice includes new technical difficulties.

Numerous research methodologies produced incorporate various types of faulty location techniques, which are divided into two categories based on how they are used: traveling

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wave-based methods and impedance-based methods [3], [4]. RES ratings are divided into three categories: small, medium, and large.

Srivastava and Parida [5] post-tripping, the exact location of the fault in any distribution line from the utility grid is predicted using a faulty locator module. This module is designed and developed by the gaussian process regression technique (GPR), and various machine learning methods used for faulty location have been explored and compared to establish the superiority of the GPR method over others.

Hussan et al. [6] point out that it has become more important in recent years for implementation of better energy management and voltage regulation in direct current microgrids (DC MG). This is particularly important for both fluctuating loads and for high-low penetration of renewable energy sources (RES). The potential of these resources has received a lot of attention due to their abundance [6].

In recent years, an in-depth mathematical model for the decentralized stochastic recursive gradient (DSRG) approach has been constructed, as stressed by Hussan et al. [7]. Recursive gradient descent updates, tie-line restrictions, and power flow constraints are all thoroughly formulated in this model. After extensive testing on a 3-area, 9-bus system, the DSRG approach showed excellent convergence speed and cost-effectiveness [7].

Recently published employ of a recommended hybrid electric vehicle (HEV) model that makes use of a fuzzy logic-based control technique for improved voltage regulation and energy management under varied load situations is covered by Hassan et al. [8]. Additionally, they compared the newly developed control system to other approaches, such as sliding mode control (SMC) and proportional-integral-derivative (PID) control [8].

In recent years implementation of machine learning (ML) methods in MG protection have attracted a great deal of attention from academics [9], [10] because of the availability and accuracy of data.

In [9], the huang hilbert transform (HHT) is applied to the non-stationary faulty current signals to extract informative features for the support vector machine and extreme learning machine (ELM) algorithms to use in the faulty event classification. Numerous signal processing techniques, including discrete wavelet transform (DWT), wavelet transform, S-transform, discrete fourier transform (DFT), and discrete orthonormal stock well Transform (DOST), have been used to preprocess the current and voltage waveforms in MG, etc. [10], [11].

However, because an additional step is involved before obtaining input signals, the feature of the extraction phase will lead of difficulty of creating (ML-based protection strategy). It needs additional equipment, such as a digital signal process (DSP) kit, to accomplish these kinds of electrical signal transformations.

Artificial neural networks have been extensively studied for their ability to recognize the presence, kind and faulty phases of a system after obtaining the input characteristics [3], [12]. Supervised learning-based methods, including support vector machines (SVM) [9], decision trees (DT) [13], K-nearest neighbor (KNN) [14], and several others have demonstrated remarkable accuracy in identifying the presence of faults and their location when used in an online manner.

Recently, numerous authors have employed various types of machine learning models to determine the exact location of a fault in a Microgrid for situations involving grid-connected and islanded mode, different fault types, fault impedance, amount of DG penetration, and load fluctuation [4], [10], [14], [15].

While load variation events increase the quantity of training data, then they have little effect on faulty location prediction [16]. As a result, since it requires less time, it proposed that the load change condition be eliminated and collecting training data [17].

The AC microgrid is easily compatible with the main grid and existing conventional home appliances, which are much more focused and widely researched [18]. renewable microgrids (RMG) are electricity distribution systems that include loads and DGs and can increase public access to DG and renewable energy [2]. However, microgrid implementation introduces new technical challenges. Many research approaches created different types of fault identification and various faulty location techniques, which are divided into two categories based on their application: impedance-based methods (low and high-impedance) and travelling wave-based methods.

Chaitanya et al. [4] use variational mode decomposition to detect high-impedance faults in a DG-based distribution system. However, these signal pretreatment strategies necessitate an additional stage of data preparation before training the ML model, raising computing overhead costs.

The potential of ML techniques to study all possible regular and aberrant operating scenarios during training motivates research to enable their widespread implementation in MG protection. These aberrant situations might occur during line failures, DG outages, unexpected load changes, and electrical device problems. As a result, ML-based schemes can protect MGs from all aberrant circumstances during operation. Conventional protective measures, on the other hand, struggle to function efficiently because of the active nature of MG operation.

The ML technique, however, necessitates a huge amount of data, storage, and retraining of ML models because the input data is updated irregularly. Data storage is possible because of the high-speed processors available in the intelligent protection system (IPS).

Our suggested technique eliminates the signal processing stage and improves execution time, allowing for faster ML retraining when needed. This study have looked into two advanced ML approaches: the ANFIS and the SVM algorithm.

Key findings include the ability of the SVM model to accurately classify fault scenarios and optimize relay coordination, minimizing fault clearance times while maintaining selectivity. Simulation results validate the robustness of the scheme under diverse fault conditions, showcasing improvements in protection speed and accuracy compared to conventional methods [19].

The above mansions' challenges to overcome required less downtime on the AC microgrid. The proposed SVM algorithm technique optimised the program execution time in the faulty location on an AC Microgrid for a grid-connected and islanded mode of operation. The proposed SVM algorithm has a minimum RMS error of 0.003 with the least execution times of **2.02 ms** for islanded mode and **2.20 ms** for grid-connected mode, which tested that it is a genuine tool for faulty location estimation. The AC MG system is well-coordinated, and the proposed approach produces excellent results in terms of program execution time for both island and grid-connected modes.

The motivation for applying the approach of the SVM algorithm includes a wide range of characteristics, such as speedy prediction, higher accuracy, faster execution time, minimization of overestimation, outliers' influence is lessened, reduced storage space, controlling high-dimensional data and improved computing complexity. The SVM algorithm estimates the probability distribution across all feasible kernel functions to fit the data set with less computing complexity. These advantages have inspired ANFIS and SVM algorithm applications in this work.

The proposed method to solve those research problems, this research method considers some research contributions, which are enlisted as follows,

- The ANFIS technique is used to propose a Fault Identification System (FIS) for detecting the precise faulted line and fault type in an AC MG in real time.
- A Faulty Location System (FLS) is also offered to achieve exact faulty location in this application using a less-studied predictive algorithm known as SVM. The SVM algorithm's benefit compared between current ML approaches used in faulty location schemes is determined.
- Proposed a novel methodology to optimize program execution time for FIS and FLS in both grid-connected and islanded modes of operation in the AC MGs for IEEE12BUS.
- From the above research gaps, to showcase the effectiveness of optimized percentage of error is compared with the recently published research articles.
- The proposed FIS and FLS protect the microgrid in both grid-connected and islanded modes of operation, enhancing its reliability.
- The proposed FIS and FLS we demonstrated through simulation in the MATLAB2023b software environment. Results are compared to optimization methods such as ANFIS and SVM algorithms in machine learning.
- Direct voltage and current measurements at source buses are employed as inputs for ML model development, minimizing the need for an additional signal processing stage.
- The proposed FIS and FLS are resilient and perfect for real-time applications as demonstrated by an accuracy investigation.

The structure of an Alternating Current (AC) MG system is shown in Figure 1. The Point of Common Coupling (PCC) is connected to the AC MG system.

In recent years, designing protection schemes were highly challenging due to the various modes of operation, i.e. grid-connected and islanded, bidirectional power flow, and the intermittent nature of RESs, which results in various levels of fault current analysis contribution as shown in Figure 1 [20].

The AC microgrid can operate in either grid-tied or island mode as well as depending on the microgrid's requirements, it can switch between the two modes smoothly or via a power interruption [21]. In islanded mode, whenever a power quality perturbation and fault occurs on the main grid, the AC MG disconnects from it and operates independently [22], [23].

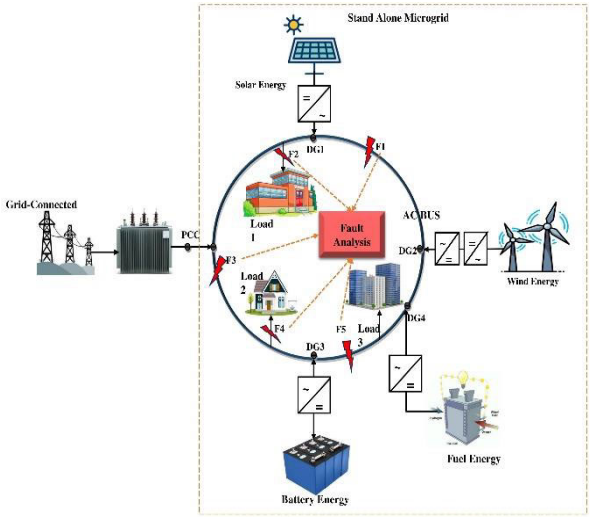

FIGURE 1. Structure of AC MG system.

In an electrical power system, the point where multiple consumers, microgrids, and loads are connected to a single power source is known as the Point of Common Coupling (PCC).

It acts as a line of demarcation between the electrical systems of the end users, Microgrid, and the utility grid. Since power quality can impacted by interactions between various systems at costumer side and the utility network, the PCC idea is crucial to power quality studies.

The coordination of fault ride-through curves and protection schemes is another issue that can impact overall microgrid stability in islanded operations [24]. Several proposals have been made to develop an appropriate strategy capable of protecting on the AC MG in both modes [25].

The proposed SVM algorithm can effectively distinguish between grid-connected and islanded modes, ensuring that the PCC operates only when necessary. You are right! One of the most important factors in ensuring the reliability and security of a Microgrid (MG) is the sensitivity of the Point of common coupling (PCC).Determining AC microgrid co.

The PCC operates as the junction between the multiple sub grids within the AC MG and the Main power grid. Faults and changes in operational modes can result in stability and operational failures because it is an important point for power exchange and synchronization.

Grid-connected and islanded modes are two types of multiple operating modes frequently observed in microgrids. The change between both these modes can be sudden and is affected by faults. The proposed SVM algorithm has been tested for fault location, protection, and control in extreme conditions, such as fast mode changes and simultaneous faults.

The proposed research article's structure is summarized as follows: In Section II, the proposed research works are explained in Model-Based IEEE12BUS AC Microgrid are presented; in Section III, the proposed research results and analysis are presented; and in Section IV, the paper concludes with future enhancements.

Nonetheless, the challenges are presented, as described in the problem statement section. So, this research methodology proposed an adaptive protection of AC Microgrid.

II. PROPOSED RESEARCH WORKS MODEL-BASED IEEE12BUS AC MICROGRID

An IEEE12BUS AC Microgrid interconnected system running at medium voltage is developed using the MATLAB SIMULINK platform. The generating the necessary dataset for training the Fault Identification (FI) and Faulty Location (FL) modules. The interconnected AC MG shown in Figure 2 is based on [26] and involves of six energy sources operating at 50 Hz.

Grid 1 (G1), Grid 2 (G2), and Grid 3 (G3) are the three main grid supplies, with three distributed generating energy sources being DGs (DG1, DG2, and DG3). The AC MG is made up of six distributed lines. CBs are installed at both ends of each line and controlled by a smart digital relay (SDR). Table 1 lists the test system's design parameters.

The main reason for using only direct voltage and current measurements as inputs for relay operating conditions lies in the practical demands of power systems and AC microgrids.

In this system, relays can operate effectively with input parameters such as direct voltage and current. The proposed model accurately finding fault identifies fault locations faults in the AC microgrid.

The input features are used to enhance the operational model accuracy of an AC microgrid. There is a need for effectiveness, real-time application, and minimizing computing overhead. However, incorporating additional elements such as harmonic analysis and derived power measurements is not warranted for this input feature.

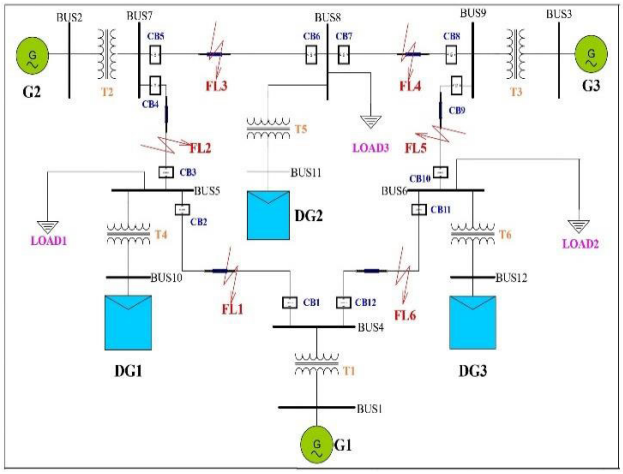

FIGURE 2. IEEE12BUS AC Microgrid test system.

TABLE 1. IEEE12BUS test system design parameters.

S No	Component	Parameters
1	Grid Sources (G1, G2 and G3)	Line Voltage 420V (11KV/0.42KV transformer), internal resistance = 0.8929Ω, internal inductance = 16.58 mH.
2	Distributed Generating energy sources (DG1, DG2 and DG3)	Line Voltage 420V (11KV/0.42KV transformer), internal resistance = 0.8929Ω, internal inductance = 16.58 mH.
3	Feeders	Line length (km) = 21.200, Positive- and zero-sequence resistance (Ohms/km) [r_1 r_0] = [2.415e-4 7.245e-4]. Positive- and zero-sequence inductances (H/km) [L_1 L_0] = [3.87e-6 11.6092e-6]. Positive- and zero-sequence capacitances (F/km) [C_1 C_0] [2.873e-6 8.619e-6].
4	Loades	Load1: P=30KW Q= 4KVAR, Load2: P=40KW Q= 5KVAR, Load3: P=20KW, Q= 2KVAR

The electrical power system's primary parameters are direct measurements of voltage, current, and frequency. The system design avoids the extra computing burden of determining secondary aspects, such as power metrics (active, reactive, and apparent power).

Real-time operation is crucial for applications such as dynamic control, fault identification, and relay protection in AC microgrids. We will consider additional input features, such as derived power metrics and harmonics analysis, in future work.

Depending on domain expertise, the decision to use direct measurements can be considered a conscious one. In systems where voltage and current directly capture the basic processes, for instance, these inputs are considerations of the system's behaviour. The design can put more emphasis

on cost-effectiveness, the convenience of utilization, and the robustness of marginal accuracy gains.

The proposed method is applicable to other systems. The proposed approach work is well-suited of AC Microgrid systems, which operate in real-time, especially in bigger systems such as the IEEE 14-Bus and IEEE 35-Bus. Addressing more complex microgrids and other power system networks requires specific adjustments.

Limitations of the Proposed Method: If the proposed method is tailored for the specific configuration of the IEEE 12-bus system, it can scale well to larger grids. Complex microgrids depending on generation and load conditions frequently display dynamic behavior.

Depending on the algorithm used (e.g., optimization, machine learning), the method may become computationally intensive as the size of the system increases and the system's overall cost will rise.

In summary, the applicability of the method to other systems and its no limitations depends on the simplification of the approach, its scalability, computational demands, and assumptions made about the microgrid's behaviour.

A. PROBLEM FORMULATION OF FAULT IDENTIFICATION ALGORITHM

The concise mathematical model is utilized to study the Root Mean Square (RMS) voltage and current data at each node. Equations (1) and (2) indicate the average deviation of the RMS voltage and current from their regular operational values for the IPS system during the startup process.

$$\Delta V_c(g) = \frac{1}{p}\sum_{i=1}^{p}\{V_c(g, y_i) - V_{cM}\} \tag{1}$$

$$\Delta I_c(g) = \frac{1}{p}\sum_{i=1}^{p}\{I_c(g, y_i) - I_{cM}\} \tag{2}$$

Here, $V_c(g, y_i)$ symbolizes the RMS voltage magnitude corresponding to the RMS c of the bus number i recorded at a particular time instant g. Here x belongs to the cyclic RMS value of (R, Y, B). The number of bus locations is being denoted by p. V_{cM} is the normal operational value of the magnitude of the RMS voltage which is taken equal to $240V * \sqrt{3} = 415V$. The symbol I_{cM} represents the electrical power of an RMS current when it's in a healthy state. However, during a fault, the current $\Delta I_c(g)$ deviates from zero for a short period of time before eventually dying out.

To determine the extent of the disturbance, we begin by calculating the Undervoltage and Overvoltage Disturbance Counters ($C^-_{v,x}$ and $C^+_{v,x}$ respectively) as specified in equations (3) and (4). These counters are dynamic parameters that remain zero when the system is in good condition. This counter activates only when there is a failure in the system. For t= 0, set $C^-_{v,x} = C^+_{v,x} = 0$, for t >0, compute $\Delta V_c(g)$ and $\Delta I_c(g)$ based on Equations (1) and (2) modify disturbance counter parameters as follows [27].

$$C^-_{v,x}(g) = \max\{C^-_{v,x}(g-1) + \mathrm{sgn}(T^-_{v,x} - \Delta V_c(g)), 0\} \tag{3}$$

$$C^+_{v,x}(g) = \max\{C^+_{v,x}(g-1) + \mathrm{sgn}(\Delta V_c(g) - T^+_{v,x}), 0\} \tag{4}$$

$T^-_{v,x}$ is the threshold upto which voltage dip is permitted during fault. If mean voltage deviation crosses this threshold, then disturbance counter $C^-_{v,x}$ gets activated and starts the counting. Faulty conditions:

A fault incipience is detected if any of the following three situations (1 to 3) is detected.

Situation 1: Single line to ground fault

$$\frac{C^-_{v,x}(g)}{F_r} + T_{lag} \geq T_{fault}. \text{ for one of the phases}$$
$$\Delta V_c(g) > 0. \text{ for remaining two phases}$$

Situation 2: Double line to ground fault

$$\frac{C^-_{v,x}(g)}{F_r} + T_{lag} \geq T_{fault}. \text{ for two of the phases}$$
$$\Delta V_c(g) > 0. \text{ for remaining one phase}$$

Situation 3: Three phase fault

$$\frac{C^-_{v,x}(g)}{F_r} + T_{lag} \geq T_{fault}. \text{ for all of the three phases}$$

wherein $C^-_{v,x}(g)$, $C^+_{v,x}(g)$ are counter for data disruption. Fr denotes the rate at which data is being sampled. The term $\frac{C^-_{v,x}(g)}{F_r}$ represents the exact time taken in the identification of fault and T_{lag} is the delay time due to processing of data through RMS and root mean voltage data Concentrators. Therefore, as soon as the value of this total time reaches the permissible fault tolerant time, the IPS detects the fault.

Only the faulted phase will see a positive value of a term $\frac{C^-_{v,x}(g)}{F_r}$ because the disturbance counter $C^-_{v,x}$ has zero value for healthy phases and positive value for faulted phases as per Equations (3) and (4).

The term V_{cM} represents the Root Mean Square (RMS) voltage, which is a measure of the effective value of a varying voltage signal. The RMS voltage for a periodic waveform is given by:

$$V_{cM} = \sqrt{\frac{1}{T}\int_0^T v^2(t)dt}$$

where:

$v(t)$: The instantaneous voltage as a function of time.

T: The period of the waveform

For standard waveforms:

Sinusoidal Voltage: If $v(t) = V_{peak}\sin(wt)$, then $V_{cM} = \frac{V_{peak}}{\sqrt{2}}$.

Square Wave Voltage: $V_{cM} = V_{peak}$ because the voltage is constant over its period.

(b) RMS Current (I_{cM}):

The symbol I_{cM} represents the RMS current, similar to the RMS voltage. Its formula is:

$$I_{cM} = \sqrt{\frac{1}{T}\int_0^T i^2(t)dt}$$

where:

$i(t)$: The instantaneous current as a function of time.

T: The period of the waveform.

For standard waveforms:

Sinusoidal Current: If $i(t) = I_{peak}\sin(wt)$, then $I_{cM} = \frac{I_{peak}}{\sqrt{2}}$

Square Wave Current: $I_{cM} = I_{peak}$.

Derivation of V_{cM} and I_{cM} for Different Operating Conditions.

The derivation of these values depends on the waveform characteristics and circuit parameters:

For Sinusoidal AC Systems: If a circuit operates under sinusoidal conditions:

Measure the peak voltage (V_{peak}) or peak current (I_{peak}).

RMS values are derived as:

$$V_{cM} = \frac{V_{peak}}{\sqrt{2}}, \quad I_{cM} = \frac{I_{peak}}{\sqrt{2}}$$

For Non-Sinusoidal or Complex Waveforms: RMS values must be computed using their definitions:

$$V_{cM} = \sqrt{\frac{1}{T}\int_0^T v^2(t)dt}, \quad I_{cM} = \sqrt{\frac{1}{T}\int_0^T i^2(t)dt}$$

Are V_{cM} and I_{cM} Standardized: In electrical engineering, RMS values are frequently utilized and standardized.

Accurate values that provide equivalent power to a DC signal can be calculated using RMS voltage and current, the standard measurements for characterizing AC waveforms. RMS values ensure precision and processing speed while allowing for comparison between multiple waveform forms.

Standards and Reference:

RMS values are defined as voltage and current by numerous electrical standards (such as IEEE and IEC) as the primary measurement parameters for AC systems. The fundamental concepts are system ratings, equipment specifications, and safety compliance with standards such as IEC, IEEE, and ANSI. Measurement tools, such as multimeters, usually require calibration in order to determine RMS values for sinusoidal signals.

B. INTELLIGENT PROTECTION SYSTEM

As shown in Figure 3, an intelligent protection system have two modeling stages: offline and online.

During the offline stage, simulation data sets are (different ML methods) collected, selected features are normalized, and FI and FL data sets are trained using the ANFIS and SVM algorithm, respectively.

The main objective of an AC microgrid's IPS is to identify faults and locate the problems inside the microgrid reliably and efficiently. Distributed energy resources (DERs) are solar panels, wind turbines, and energy storage devices that are part of complex power systems known as AC microgrids. The grid-connected and islanded modes of operation present special fault management challenges for these systems. To address these issues, an IPS uses the latest technologies, including real-time analytics, machine learning, and artificial intelligence.

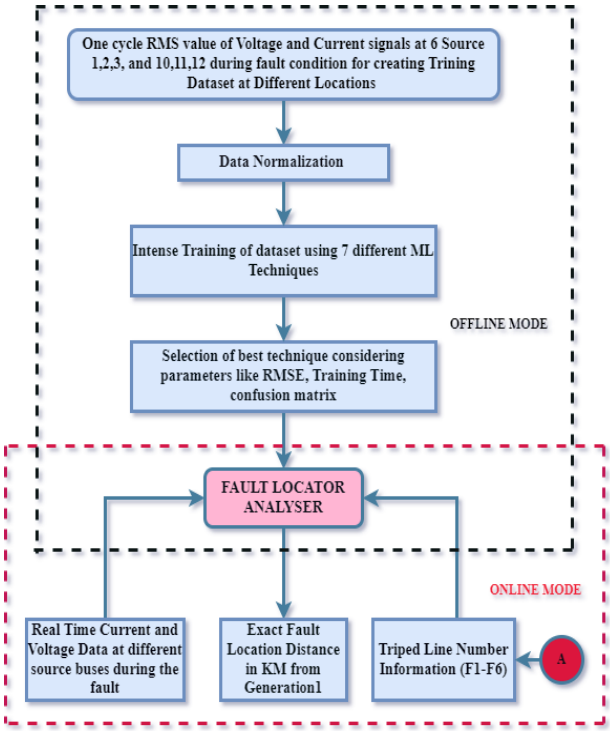

FIGURE 3. Flow chart of intelligent protection system.

Once trained, the AC microgrid is ready to utilize in an online application. Real-time field data collected by the IPS is used to identify the type and location of the fault. The RMS value of postfault voltage and current measurements from six source buses are utilized as an input feature for test system development. To design an effective and long-lasting test system.

The test system must be trained on data containing information about all conceivable AC microgrid operation conditions. Changes in DG penetration and load are also included in the data sets produced to illustrate AC MG operation. The measurement of three-phase voltages and currents from six-generation buses produces 36 input features (6 × 6).

An AC Microgrid system consists of multiple components such as generators, inverters, batteries and load profiles. Consequently, the input features can encompass a wide range of variables that characterize the current status of the electrical grid. The selection of input features depends on the supply-generating system. The number of cases in the grid-connected mode is 62208 (18 × 4 × 3 × 3 × 24 × 4), and the number of cases in the islanded mode is also 62208 (18 × 4 × 3 × 3 × 24 × 4). The above Figures 2. do not account for the distributed generation (DG) penetration level.

Therefore, the total number of simulated cases is 124416 (62208 + 62208), which can be calculated as follows $(36 \times 4 \times 3 \times 3 \times 24 \times 4) = 124416$. The dataset for the AC Microgrid is expected to include various types of sensor readings and operational data points, such as voltage, current, power consumption, and renewable energy output.

Feature selection is a crucial step in preparing the data for training machine learning algorithms for an AC Microgrid, as the numerous input features may contain noisy and redundant information. As the number of selected input features increases, leads to a forementioned. By reducing the size of the dataset with the most important underlying patterns, dimensionality reduction can help to overcome these problems.

The fault distances from G1 in each line are as follows. Line 1 is 0-15 kilometers from G1; Line 2 is 10-25 kilometers; Line 3 is 20-35 kilometers; and Line 4 is 30-40 kilometers from G1. Line 5 is 20-30 kilometers from G1, whereas Line 6 is 0-10 kilometers from G1. Data is pre-processed before data is provided to the machine learning test system for training. During a single fault cycle, we solely collect RMS voltage and current measurements. This stage is critical to collecting system data, as certain data may be missing and excessive. In addition, feature values vary significantly. To improve the accuracy of output prediction, all features should be within a specific range. Chahal and Gulia [28] suggest using the standardization approach for (input c) to achieve the goal.

$$c' = \frac{\text{c-mean(c)}}{\text{standard-deviation(c)}} \tag{5}$$

The research approach focuses on practical grid scenarios to improve the accuracy of the trained test system. The trained test system is tested with new data to ensure its reliability. Figure 2 shows that the FI and FL are trained with 36 input features and 3456 observations. After designing the test systems, they can be integrated into IPS. When a fault occurs in online mode, the IPS examines voltage and current data from all source buses before sending it to the FI and FL.

FI will identify the fault type and affected line number (lines 1-8). The IPS will issue a trip command to the CBs of the faulted line. In addition, the IPS will require a trip command to the CBs on the affected line. After isolating the faulty line, the FL indicates the exact location of the fault on G1. The below sections provide an in-depth discussion of the ML approaches ANFIS and SVM algorithms.

C. FAULT IDENTIFICATION AND FAULT LOCATION BY ANFIS AND SVM APPROACHES

SVM and ANFIS approaches are employed to detect and locate faults in the modelling of AC MG. The ANFIS algorithm identifies the type of fault, while the SVM algorithm calculates the fault distance.

A fault identification system (FIS) is designed to detect and classify the types of problems that occur in an AC microgrid system. Its main objective is to quickly identify any abnormal condition affecting the AC Microgrid system's functionality, such as open circuits, short circuits, and equipment failures.

Finding the exact fault location system (FLS) of a problem in an AC Microgrid system is the next step for a fault location system. This solution is essential for minimizing maintenance costs and downtime as it permits rapid restoration and repair.

The Adaptive Neuro-Fuzzy Inference System (ANFIS) is a hybrid intelligence system that analyzes complex nonlinear systems by combining the advantages of neural networks and fuzzy logic. ANFIS integrates the interpretability and decision-making capabilities of fuzzy systems with the learning capacity of neural networks.

1) ADAPTIVE NEURO-FUZZY INFERENCE SYSTEM (ANFIS) APPROACH

An Adaptive Neuro-Fuzzy Inference System is a model that uses several components to infer relationships between input and output variables. It consists of different layers that work together to process input data and generate output predictions.

The input layer contains nodes that represent the input variables of the system, each node representing a linguistic variable or a feature of the input data. These nodes encode the input information and pass it to the subsequent layers.

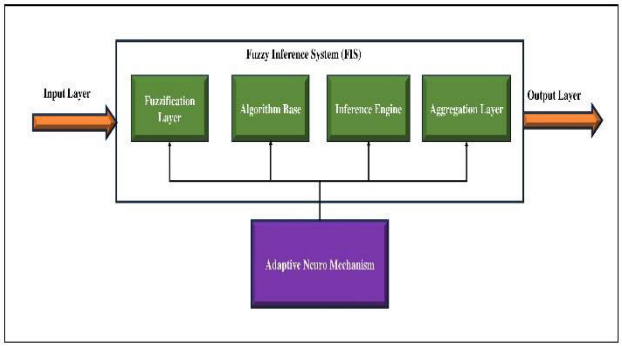

FIGURE 4. Architecture of adaptive Neuro-Fuzzy inference system.

Figure 4 illustrates the fuzzy inference system (FIS) is the core of ANFIS and comprises fuzzy logic components. It consists of several layers, including the fuzzification layer, algorithm base, fnference engine and aggregation layer.

- The fuzzification layer maps the crisp input data into fuzzy linguistic terms and each input node is associated with membership functions that define the degree of membership of the input variable to each linguistic term.
- The algorithm base contains a set of fuzzy if-then rules that define the relationship between the input variables and the output variable. Each algorithm specifies a combination of linguistic terms from the input variables and the associated output.
- The inference engine computes the degree of activation of each algorithm based on the input values and the corresponding membership functions. It combines the fuzzy algorithm to generate fuzzy outputs.
- The aggregation layer aggregates the fuzzy outputs from different algorithm to obtain a single fuzzy output that

represents the overall system response. The adaptive mechanism of ANFIS adjusts the parameters of the fuzzy inference system based on the training data.

Fault identification in an AC microgrid is to identify the types of faults and defects (open circuits, short circuits, grounding faults, and load imbalances) that interfere with the system's regular operation. Due to the dynamic nature of AC signals and the integration of renewable energy, these faults frequently display complicated, nonlinear, and time-varying behaviours. ANFIS is particularly well-suited for this task for the following reasons:

- ANFIS is an excellent choice for modeling the intricate behaviour of defects since it combines fuzzy logic with neural networks, which can handle such nonlinear relationships.
- ANFIS can manage this uncertainty by classifying fault types using fuzzy inference techniques, assisting operators in accurately identifying problems.
- Due to fluctuating loads, renewable energy sources, and changing operational conditions (grid-connected vs. islanded mode), microgrids are by nature dynamic.
- By learning from past data and continuously modifying its model to detect problems under various settings, ANFIS can adjust to these changes.
- Because of its flexibility, ANFIS is especially useful in microgrids where load variations and fluctuations in renewable energy sources can lead fault conditions to vary.

This adaptation enables ANFIS to learn and improve its performance over time. The adaptation process typically involves techniques such as gradient descent or least squares estimation to update parameters. The output layer consists of a single node that computes the crisp output of the ANFIS model based on the aggregated fuzzy output obtained from the FIS. This node performs defuzzification to convert the fuzzy output into a crisp value that represents the system's prediction or inference.

ANFIS, which stands for Adaptive Neuro-Fuzzy Inference System, combines neural networks and fuzzy logic. It utilizes fuzzy rules to emulate the knowledge of human experts regarding the behaviour of a system, and neural network learning techniques are employed to adjust these fuzzy rules based on input-output data. ANFIS is particularly beneficial in fault identification systems, especially where measurement results can be affected by errors. It proves effective when dealing with imprecise and unclear data. Further, ANFIS excels in modelling systems that are challenging to represent using conventional linear techniques, as it can capture nonlinear interactions between inputs and outputs.

The supervised machine learning approach known as support vector machine performs exceptionally well in classifying data into multiple categories. It achieved by identifying a hyperplane in the feature space that maximizes the margin between different classes. SVM is effective in high-dimensional spaces and delivers excellent results when working with large sets of features, such as sensor data, as well as when detecting intricate and subtle fault patterns. Moreover, SVM can handle non-linear data separability by transforming the data into higher dimensions using kernel functions. It is particularly recognized for its strong generalization capabilities, even when dealing with smaller datasets, especially in scenarios where the differentiation between faulty and normal states is clear.

Fault localization and identification regularly involve both clear-cut and complex situations. Because a single model struggles to handle all categories simultaneously, some defects have worse fault identification accuracy. While support vector machines excel at classifying separable data, ANFIS are particularly effective for modelling fuzzy and uncertain systems. A hybrid and generalized model lacks the flexibility to apply different solutions based on the type of defect, which can hinder its effectiveness in balancing both characteristics. Although a single generic countercontrol can simplify fault identification in certain scenarios, it reduces accuracy when dealing with diverse and complex fault types.

SVM (which is effective for classification in high-dimensional spaces) and ANFIS (which manages uncertainty and fuzziness) are strengthening. In some cases, integrating them into a single system can lead to a loss of performance and specialization. Therefore, it produces better results to maintain separate models and employ a hybrid strategy that dynamically selects the optimal method—either ANFIS and SVM—based on the specific features of the fault identification and placement problem rather than attempting to combine them into a single model.

Adaptive neuro-fuzzy inference systems must be able to adjust to significant variations in distributed generation (DG) and load in an AC microgrid. Here are some details on the potential constraints of ANFIS and how it can handle such variations:

All possible practical distribution generations and various modes of operation (such as grid-connected and islanded) are taken into consideration while selecting the data set. The IEEE12BUS AC Microgrid model that has been developed is sufficiently resilient for operating in any kind of environment.

The training dataset's quality and diversity have an enormous impact on how effectively ANFIS performs.

When generating the dataset, researchers analysed the IEEE12BUS AC Microgrid system for every potential distribution generation, load conditions, and operating mode in many different situations.

- Nonlinear relationships between input variables (such as load demand and DG output) and control outputs (e.g. voltage and current) can be learned by utilizing ANFIS.
- By enhancing its membership functions and rules through training, it may adjust to changing situations.
- Variable renewable energy generation and dynamic load variations are two examples of the uncertainties and nonlinearities that ANFIS can manage.
- ANFIS can maintain system stability and improve performance in rapidly changing situations when trained properly.

The adaptive neuro-fuzzy inference system is a hybrid intelligence system that analyzes complex nonlinear systems by combining the advantages of neural networks and fuzzy logic. ANFIS integrates the interpretability and decision-making capabilities of fuzzy systems with the learning capacity of neural networks.

The integration of ANFIS with powerfully designed feature extraction algorithms that gives it its unique. By analyzing IEEE12BUS AC Microgrid electrical parameters such as voltage, current, and impedance, the system can effectively detect and classify faults, even under the dynamic and nonlinear operating conditions typical of AC microgrids. Estimation of symmetrical and unsymmetrical faults can be done in IEEE12BUS AC Microgrid system by utilizing ANFIS techniques. The application of ANFIS enhances the reliability and safety of microgrid operations by providing precise fault identification and location capabilities.

Advanced implementations can involve ANFIS in real-time decision-making systems for microgrid protection. This includes detecting high-impedance faults, islanding conditions, and abnormal power quality, triggering control actions such as relay coordination and isolation.

In summary, while ANFIS itself is an established technology, its application in fault identification and location within AC microgrids represents a significant innovation. By combining ANFIS with specialized data processing and feature extraction techniques, these systems can effectively address the unique challenges of microgrid protection.

2) SVM APPROACH

From the selected optimum relay time, the relay time is estimated using SVM approach. The model aims to draw decision boundaries between data points from different classes and separate them with maximum margin. The reason for choosing the SVM approach is,

a) High-dimensional spaces had more benefits than SVM.

b) Also, it handles both classification and regression on linear and non-linear data and finds the complex relationship between the inputs. This is because many relay operating time is presented in the adaptive protection process of the microgrid.

c) The SVM will perform when the number of features for each data point is greater than the number of training data samples.

d) Since the support vector classifier places data points above and below the classification hyperplane, there is no probabilistic justification for the classification.

e) SVM performs excellently when there is a significant gap between classes and when there are more dimensions than samples.

f) Diverse kernel functions can be offered for the decision function, making it versatile.

The structure for SVM is shown in Figure 5,

The data given for classifying is represented as a unique point in a space where each point is represented by some vector u.

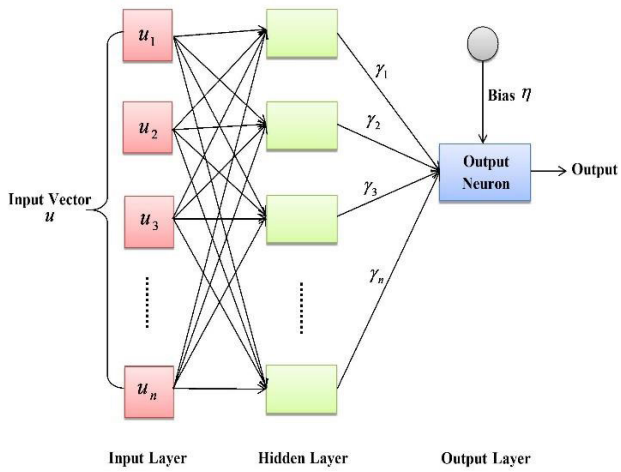

FIGURE 5. Structure for SVM.

$$u_i \in Q^n \tag{6}$$

Here, Q^n specifies a vector space with n dimension. Another vector value is described in equation (7),

$$v \in Q^l \tag{7}$$

where, l specifies the vector space with l dimension. For constructing an optimal separating hyperplane:

$$\min \frac{1}{2} a^T . a + D \sum \delta_i \tag{8}$$

$$\text{Subject to, } v_i \left(a^T . u_i + \eta\right) \geq 1 - \delta_i \tag{9}$$

$$\delta_i \geq 0, \ \ i = 1, \ldots.., l \tag{10}$$

where, δ_i is an additional variable that characterizes the magnitude of the error u_i belonging to the training set, D is the parameter of the algorithm, and l indicates the number of classes. Then, the training of the data class is defined as in equation (11),

$$\min \frac{1}{2} \gamma^T . Z\gamma - e^T \gamma \tag{11}$$

$$\text{Subject to, } v^T \gamma = 0 \tag{12}$$

$$0 \leq \delta_i \leq D, \ \ i = 1, \ldots\ldots, l \tag{13}$$

where, γ defines a dual variable, e^T is the vector of all ones, the parameter $D > 0$ is the upper bound and Z is a l positive semi-definite matrix. Thus, the matrix is derived as follows,

$$Z_{ij} = v_i v_j u_i^T u_j \tag{14}$$

The decision function is derived by the relation of the weight function, which is derived in equation (15), and the decision function is given in equation (16),

$$a = \sum v\gamma u \tag{15}$$

$$\text{sgn}\left(\sum v\gamma u + \eta\right) \tag{16}$$

Input: Optimum relay time
Output: estimated relay time

Begin
  Initialize vector u, additional variable δ_i, and weight a
  For each u **do**
    Construct hyperplane
    if $v_i\left(a^T . u_i + \eta\right) \geq 1 - \delta_i$ **{**

$$\min \frac{1}{2} a^T . a + D \sum \delta_i$$

    } else {
      Take iteration
    } end if
    Construct decision function sgn ()
    if $loss = satisfied$ **{**
      Return
    } else {
      Iteration $it = it + 1$
    } end if
  End for
End

where, u is a vector and it is categorized as $\eta = a^T u_k - v_k$. In this, $\gamma_k = 0$.

Finding the exact location and grid segment where the fault occurred is a crucial next step after finding a failure in the AC microgrid. For appropriate fault isolation and to reduce downtime, fault location is essential. The SVM algorithm is particularly effective for this task due to the following reasons:

The objective of the SVM algorithm is to determine the best decision boundary that maximizes the group margin.

Classifying faults into different regions and sections of the microgrid (such as a specific distribution line and bus) is frequently the challenge when it comes to fault location.

SVM works well for this reason because it can clearly define the limits of choice between different fault locations based on sensor data (such as voltage and current measurements collected by different grid nodes).

Even with complex and nonlinear fault location data, the kernel method in SVM allows these high-dimensional features to be transformed into a higher-dimensional space that facilitates fault classification.

By building several binary classifiers, each specific to identifying the fault's location (e.g., which line and bus the fault is on), the SVM algorithm can effortlessly handle such multiclass problems.

The pseudocode for the SVM approach is given above. It gathers all input and target sample data, as well as sampling information for various generating variables (also known as input sample data) and faulty location lines on the variables affecting fault currents. After that, the further process is carried out, and the performance is analyzed in the further section.

These power electronics switches will have a short-time withstand capability in a millisecond range. The AC Microgrid requires instant fault isolation within milliseconds. The protection device needs maintenance due to its sensitivity and faster response. The operational and program execution times of the power system and AC Microgrid's system protective devices can be operated in milliseconds or microseconds. The faculty clearing time is needed in milliseconds for both grid-connected and islanded modes on the AC Microgrid. These factors have led to the proposed research method, utilizing the proposed SVM algorithm approach to focus on the system's capacity to operate at a suitable time.

Highlighted the importance of AC microgrids to modern distributed power systems, particularly for facilitating the integration of renewable energy sources, enhancing reliability, and reducing energy expenses. Emphasize how Support Vector Machines are currently utilized in microgrids with operations involving fault location, stability analysis, load forecasts, and power quality enhancement. The results show that the proposed approach beats conventional SVM algorithm implementations' speed, accuracy, and robustness for particular AC microgrid demands.

The benefits of the widespread utilization of the SVM algorithm of AC microgrid operation and current methods struggle with real-time implementation, flexibility, and managing dynamic operational conditions. The proposed SVM technique involves a dynamic kernel function designed specifically to deal with stochastic and nonlinear characteristics of AC microgrids.

With error rates of 0.0% and 0.183%, this method improves fault location and identification accuracy while minimizing execution time to 2.02 milliseconds compared to conventional models. We have demonstrated its scalability, resilience, and adaptability by testing real-time information, resulting in a useful tool for modern AC microgrid management systems. In addition to addressing current issues, the proposed SVM algorithm research project provides the method for more sustainable and effective energy systems.

In situations where there are more dimensions (features) than samples, SVM performs exceptionally well. Finding the hyperplane that optimizes the margin between classes is the main goal of the procedure, which guarantees a strong classification border. When dealing with non-linear data, SVM employs the "kernel trick" to translate it to higher dimensions where it can be separated linearly. This adaptability enables the proposed SVM techniques to work with different kinds of data. Using the regularization parameters voltage, current, and impedances, one may manage the trade-off between optimizing the margin and lowering error on training data.

The proposed SVM method is useful for application, data classification, and fault location because it avoids the curse of dimensionality and performs well with sparse data. The SVM techniques based on convex optimization concepts ensure

that, unlike techniques that could converge to local minima, a globally optimal solution will be found (for specific kernels and situations). SVM performs effectively with linearly separable data when using the linear kernel and with advanced kernels for non-linear data. This application helps to avoid overfitting, especially with noisy datasets.

A lot of machine learning models, particularly neural networks, are opaque. It can be challenging to comprehend their decision-making process, which can be problematic in crucial fields such as law and healthcare. While complex models might be effective in training data, they might not generalize to new, untested data. Applications of machine learning in data mining and surveillance may violate people's privacy. ML model integration into current systems can be challenging and necessitate extensive re-engineering. Regular retraining and updates are necessary because models may deteriorate over time due to concept drift and changes in data distributions.

A key element of various machine learning techniques, including support vector machines and kernel-based learning methods, are kernel functions. Selecting the right kernel function and optimizing its parameters is crucial for accurate and reliable fault location results. A common option for fault location is the radial basis function (RBF) kernel due to its effective modeling of non-linear relationships. This technique employs distance to compute feature space similarity and the accuracy of the kernel's fault location estimation during validation.

The RBF conditions and polynomial kernels are more suitable for non-linear complex situations, while a linear kernel is proven effective for simpler cases. The accuracy of fault location is greatly impacted by the kernel choice and its parameters. Because of their flexibility in non-linear situations, RBF kernels are frequently used; however, domain-specific data and performance measures should always be the basis for selection. Viability concerning resource utilization and execution, particularly in real-time systems.

D. INTELLIGENT PROTECTION SYSTEMS FOR IEEE12BUS AC MICROGRID

The IPS in an IEEE12BUS AC microgrid, which is designed to ensure system reliability and stability by detecting and localizing faults, disturbances, and abnormalities [29].

The proposed approach systems utilize advanced technologies involving the adaptive neuro-fuzzy inference system and the machine learning SVM algorithm to improve the effectiveness of protection mechanisms.

The protection system can dynamically adjust its settings in response to real-time operating conditions and changes in the system topology, ensuring optimal protection coordination and fault identification as shown in Figure 6. ANFIS algorithms can quickly detect and classify faults in the microgrid, distinguishing between different fault types such as line-to-ground, line-to-line, and three-phase. Intelligent

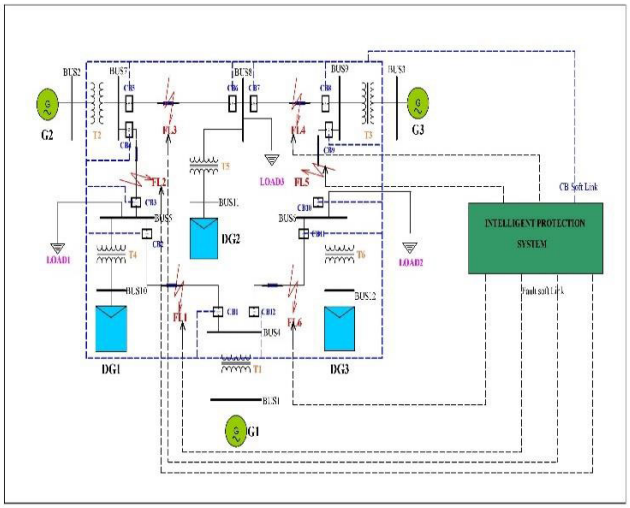

FIGURE 6. Intelligent protection systems for IEEE12BUS AC microgrid depicts intelligent protection systems.

protection systems can detect islanding conditions and execute appropriate control actions to ensure the stability and safety of the microgrid.

With the increasing integration of digital communication and control technologies in microgrids, intelligent protection systems must include cybersecurity measures for protection against cyber threats. Intelligent protection systems can detect possible equipment failures before they happen, reducing interruptions and maintenance costs.

The IEEE12BUS AC Microgrid system has 24 relays, only the relay nearest to the fault will trip. The efficiency of a relay depends upon how fast and accurately it reacts to a nearby failure. This configuration lacks redundancy; only the closest relay will function, ensuring fault tripping occurs within the allotted time.

In AC Microgrid systems, the IPS is designed to improve reliability and accuracy by ensuring that only the faulted sections of a network are isolated while unaffected sections remain operational. distributed energy resources and decentralized AC microgrids need sophisticated protection plans adapted to their operating conditions.

Digital protective relays (DPRs) and other high-speed, high-resolution sensors are used by the IPS to continuously monitor network parameters such as voltage, current, and phase angles. The proposed methods to determine the exact location of the fault consist of directional comparison, distance protection, differential protection, and overcurrent protection. The result reduces the risk of under-tripping (leaving a fault active) and over-tripping (affecting nearby lines). The network is separated into protection zones by the IPS.

Modern IPS uses protocols such as IEC 61850 and high-speed fiber optic communication to share fault information instantly between relays and control centers. These ensure that adjacent relays verify fault data before issuing trip commands, preventing unnecessary tripping.

Each critical section of the network has redundant protective relays (primary and backup). If one relay fails and sends a false trip signal, the backup verifies the fault before acting.

The system adjusts thresholds and logic dynamically based on network conditions, reducing the likelihood of false positives.

Relays are programmed to trip conservatively only when multiple independent signals confirm a fault.

The system cross-check's fault data from multiple sensors along the affected line and adjacent sections to validate the fault's existence and location.

III. RESULTS IN DISCUSSION FOR AN IEEE12BUS AC MICROGRID

In this section, the performance of the proposed research is investigated, and the research is implemented in the working platform of MATLAB Simulink.

A. PERFORMANCE ANALYSIS

The performance of the approach is analyzed for two operational phases: islanded mode (when the microgrid is disconnected from the main utility but remains energized and continues to provide power to local loads) and grid-connected mode (when the microgrid can exchange power with the external grid to ensure a steady supply within the local microgrid). We will discuss these modes in more detail in the following section.

1) ISLANDED MODE OF OPERATION

This subsection segments the performance analysis into four sections: (a) Fault Identification (FI) using the proposed method ANFIS (fault identification and different types of faults), (b) Comparison of error percentages on AC Microgrid islanded mode of operation using various machine learning approaches, (c) Applying Different Machine Learning Approaches to predict the location of faults in Islanded mode, (d) Estimation of Percentage Error, and (e) Evaluation of Program Execution Time with Different ML approaches.

a: FAULT IDENTIFICATION ANALYSED USING THE PROPOSED METHOD ANFIS (FAULT IDENTIFICATION AND DIFFERENT TYPES OF FAULTS)

Figure 7 illustrates that it is critical to identify several fault lines in the AC microgrid. Faults can occur due to various reasons such as unsymmetrical faults, short circuits and line faults.

Determine the types of faults that can occur such as line-to-line faults, line-to-ground faults, or three-phase faults.

The ANFIS is very beneficial for analyzing complicated, nonlinear systems and making inference-driven judgments.

The proposed ANFIS algorithm considers all types of faults and can quickly detect and classify faults in the microgrid, distinguishing between symmetrical and unsymmetrical faults. The IEEE12BUS AC Microgrid system is designed to analyse symmetrical and unsymmetrical faults. Two types of symmetric faults are line to line to line to ground (LLLG) and line to line to line (LLL). Three types of unsymmetrical faults are line-to-ground (LG), line-to-line (LL), and line-to-line-to-ground (LLG). Unsymmetrical faults are less severe but controlled properly to avoid power quality loss, while symmetrical faults are more serious and demand immediate action.

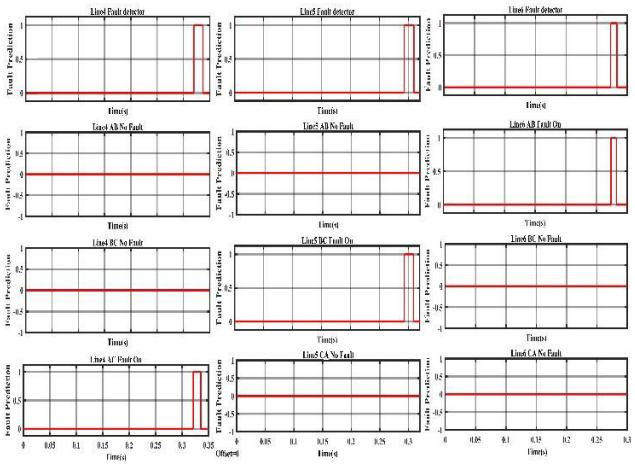

FIGURE 7. Analyzed different lines for fault identification.

In AC Microgrid systems, line-to-ground faults are the most common type of fault encountered, and implementing them is a significant focus of research and analysis. Because these faults have a substantial practical impact, studies often prioritize their detection and mitigation. In some cases, if a method can automatically detect and respond to line-to-line faults, including all fault types in the analysis may complicate the study without adding much value. Therefore, by concentrating on the most prevalent and serious fault types, research can be streamlined and clarity can be enhanced.

The proposed method is flexible and capable of addressing various types of faults, including line-to-line issues, while emphasizing the most critical problems. For example, the AC Microgrid (ANFIS and the SVM algorithm) are designed to detect deviations from normal behaviour and could potentially identify line-to-line issues even if they have not been specifically studied.

Here's how ANFIS can be used to identify various types of faults in an IEEE12BUS AC microgrid in islanded mode. First, membership functions are defined for input variables such as voltage, current, etc. These membership functions represent linguistic variables such as "low," "medium," and "high" to describe the state of each variable.

Membership functions are utilized to detect faults by capturing both normal operation and various fault conditions. The ANFIS model parameters, such as membership function parameters and rule weights are improved using a hybrid learning approach that incorporates gradient descent and least squares techniques. Training data, which consists of input-output pairs (measurements of system variables and related fault labels) is used to adaptively alter the test system parameters to reduce the error between expected and actual outputs.

After training, the ANFIS test system can be used to detect faults in real-time operations or large systems. Measurements of system variables such as voltage and current are fed into the trained ANFIS test system as input. The test system employs fuzzy logic inference based on the given membership functions and approaches to classify the system's current state as normal or faulty and identify the type of fault if present. The output of the ANFIS test system provides information about the likelihood and type of fault Identification, allowing for timely intervention and corrective action. The performance of the ANFIS-based fault identification system is evaluated using simulated or experimental data representing various fault scenarios.

In microgrids, fault identification and location are crucial to protecting the system's reliability and minimizing severe failures. The proposed strategy it must be will focus on fast and accurate fault identification can enhance fault tolerance by facilitating quicker isolation of the problem region and reducing its impact on the rest of the system. Renewable energy sources (RESs) are frequently used in AC microgrids, and their fluctuation can make fault management more difficult. According to a recent publication, "Smooth and Uninterrupted Operation of Standalone DC Microgrid Under High and Low Penetration of RESs," fault identification systems can manage sudden changes in RES power generation while maintaining balance.

Decentralized fault identification is crucial in large microgrids and multi-area power systems. A framework for decentralized optimization in large systems is provided in the following paper, "Decentralized Stochastic Recursive Gradient Method for Fully Decentralized OPF in Multi-Area Power Systems," which can be used for microgrid fault control. The recently published articles on the above mentions have been thoroughly read, focusing on the challenges of sudden changes in RES power system generation and large microgrid, multi-area power systems.

The above mentions' challenges are to overcome required less downtime on the AC microgrid. The proposed SVM algorithm technique optimised the program execution time in the faulty location on an AC Microgrid for a grid-connected and islanded mode of operation. The proposed SVM algorithm has a minimum RMS error of 0.003 with the least execution times of 2.02 ms for islanded mode and 2.20 ms for grid-connected mode, which tested that it is a genuine tool for faulty location estimation.

The AC MG system is well-coordinated, and the proposed approach produces excellent results in terms of program execution time for both island and grid-connected modes.

b: FAULTY LOCATION (FL) ANALYSED USING THE PROPOSED APPROACH SVM ALGORITHM

The estimated proposed SVM algorithm percentage error with different ML approaches for the AC MG islanded mode of operation is shown in Table 2. The lower and upper percentage error limits are 0.021 and 0.167, respectively. The SVM method faulty location is less than the compared GRNN, NN, GPR, DT, BT and ensemble for every positive percentage error value specified. The percentage of error negative value indicates that the SVM approach faulty distance is longer than the SVM technique.

TABLE 2. Comparison of error percentages on AC MG island mode of operation using different ML approaches.

Data Points	Actual Faulty Location (Km)	SVM %Error	GRNN %Error	NN %Error	GPR %Error	DT %Error	BT %Error	EN %Error
1	6.000	-0.133	0.250	-1.889	-0.300	-1.667	-0.709	-1.955
2	15.840	0.025	-0.505	-1.978	-0.114	-0.631	-0.268	-0.741
3	26.400	0.023	-0.189	-1.187	-0.068	-0.379	-0.161	-0.444
4	30.000	-0.167	0.333	-1.044	-0.060	-0.333	-0.142	-0.391
5	9.000	-0.078	0.256	-3.481	-0.200	-1.111	-0.473	-1.303
6	19.360	0.021	-1.033	-1.618	-0.093	-0.517	-0.220	-0.606
7	12.920	-0.263	0.619	-2.425	-0.139	-0.774	-0.329	-0.908
8	22.000	0.023	0.000	-1.424	-0.082	-0.455	-0.193	-0.533
9	7.040	0.028	-0.355	-1.610	-0.256	-1.420	-0.604	-1.666
10	15.840	0.025	-1.263	-1.978	-0.114	-0.631	-0.268	-0.741
11	22.500	-0.076	-2.222	-1.393	-0.080	-0.444	-0.189	-0.521
12	35.200	0.023	1.108	-0.890	-0.051	-0.284	-0.121	-0.333
13	10.640	-0.263	-1.974	-2.945	-0.169	-0.940	-0.400	-1.103
14	16.500	-0.055	1.515	-1.899	-0.109	-0.606	-0.258	-0.711
15	24.640	0.024	-1.583	-1.272	-0.073	-0.406	-0.173	-0.476
16	31.680	0.022	0.789	-0.989	-0.057	-0.316	-0.134	-0.370

The GPR technique is the lowest percentage of error analyzed with boundary lower and upper percentage error limits of 0 to 2.222, while the NN approach is the highest percentage of error analyzed with boundary lower and upper percentage error limits of 0.890 to 3.481.

The proposed SVM algorithm approaches the lowest and maximum percentages of error limitations of 0.021 and 0.167 respectively, when compared to all different ML approaches with the lowest percentage of error. The proposed SVM algorithm strategies are the most suitable approach. It is optimized for faulty locations using the SVM technique.

c: APPLYING DIFFERENT MACHINE LEARNING APPROACHES TO PREDICT THE LOCATION OF FAULTS IN ISLANDED MODE

The comparison of different machine learning approaches to predict the location of faults in islanded mode for better clarification and faulty location prediction concerning the actual location is plotted in Figure 8.

In Table 3 and Figure 8, the proposed SVM algorithm is estimate the faulty location most accurately for the Islanded mode of the AC MG system. GRNN, DT, GPR, BT, and ensemble prediction are also closer while NN is given the

TABLE 3. Comparison of faulty location on AC MG island mode of operation using different ML approaches.

SNO	Actual Faulty Location (Km)	SVM	GRNN	NN	GPR	DT	BT	Ensemble
1	6.000	5.992	6.015	5.887	5.982	5.900	5.95747	5.88269
2	15.840	15.844	15.760	15.527	15.822	15.740	15.79747	15.72269
3	26.400	26.406	26.350	26.087	26.382	26.300	26.35747	26.28269
4	30.000	29.950	30.100	29.687	29.982	29.900	29.95747	29.88269
5	9.000	8.993	9.023	8.687	8.982	8.900	8.95747	8.88269
6	19.360	19.364	19.160	19.047	19.342	19.260	19.31747	19.24269
7	12.920	12.886	13.000	12.607	12.902	12.820	12.87747	12.80269
8	22.000	22.005	22.000	21.687	21.982	21.900	21.95747	21.88269
9	7.040	7.042	7.015	6.927	7.022	6.940	6.99747	6.92269
10	15.840	15.844	15.640	15.527	15.822	15.740	15.79747	15.72269
11	22.500	22.483	22.000	22.187	22.482	22.400	22.45747	22.38269
12	35.200	35.208	35.590	34.887	35.182	35.100	35.15747	35.08269
13	10.640	10.612	10.430	10.327	10.622	10.540	10.59747	10.52269
14	16.500	16.491	16.750	16.187	16.482	16.400	16.45747	16.38269
15	24.640	24.646	24.250	24.327	24.622	24.540	24.59747	24.52269
16	31.680	31.687	31.930	31.367	31.662	31.580	31.63747	31.56269

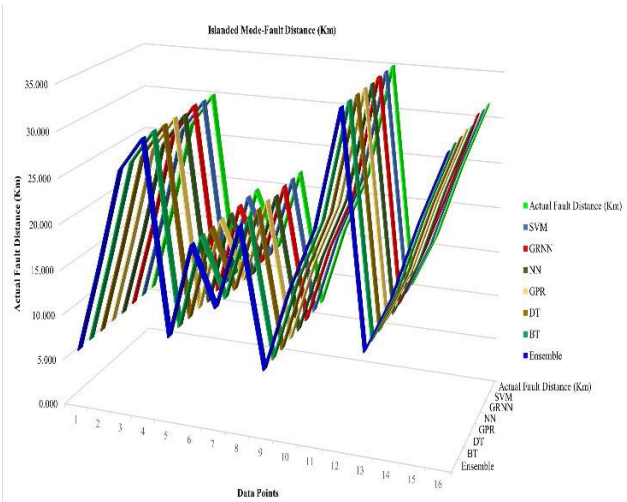

FIGURE 8. Comparison of different machine learning approaches to predict the location of faults in islanded mode.

worst results. The error observed in the NN method is very high up to 0.890% and 3.481%.

d: ESTIMATION OF PERCENTAGE ERROR

The estimated percentage error of the proposed SVM algorithm for the AC MG islanded mode of operation is shown in Figure 9, which employs various machine-learning

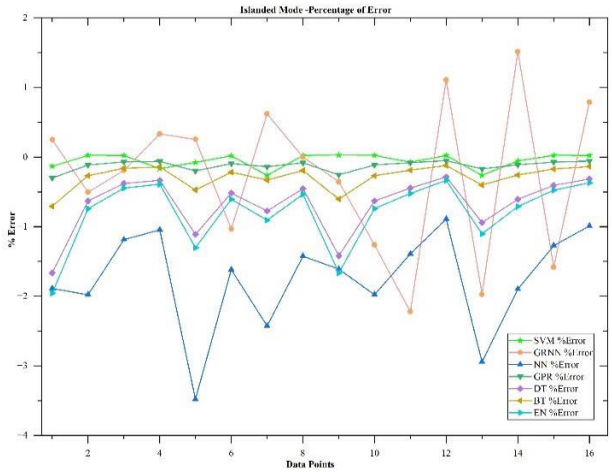

FIGURE 9. Estimation of percentage error graph Islanded mode.

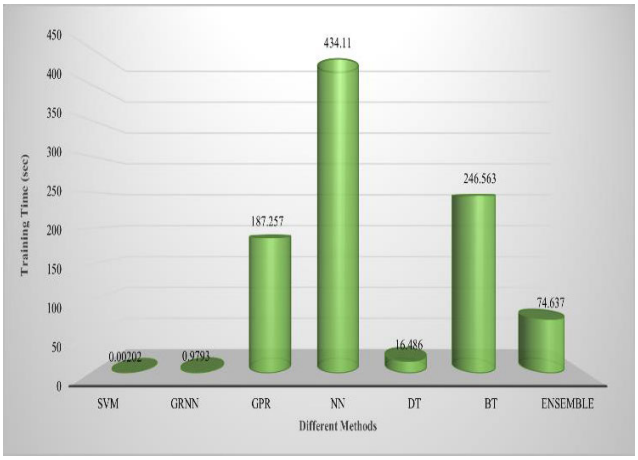

FIGURE 10. Evaluation of program execution time with different ML approaches islanded mode.

approaches. The maximum and minimum percentage error limits are 0.167 and 0.021 in that arrangement. The GRNN, NN, GPR, DT, BT and ensemble perform worse than the SVM technique for each positive percentage error number provided.

The SVM method's fault distance is greater than the SVM algorithm methods, as indicated by the negative percentage of error value. The SVM approach is employed to optimize faulty locations.

e: EVALUATION OF PROGRAM EXECUTION TIME WITH DIFFERENT ML APPROACHES

Figure 10 and Table 4 depicts the implementation of the proposed SVM algorithm approach to estimate the program execution time at the faulty location of an AC Microgrid for an islanded mode of operation. The lowest percentage of errors is determined by the NN method, although program execution time (434.11s) takes longer. Because of the extended program execution time, the AC MG system is not in synchronization. The AC MG system the entire system will

TABLE 4. Comparison of training time on AC MG island mode of operation using different ML approaches.

Data Points	Islanded Mode	Proposed Approach SVM Algorithm	ML Techniques Tools					
			GRNN	NN	GPR	DT	BT	Ensemble
1	Training Time (S)	**0.00202**	0.9793	434.11	187.257	16.486	246.563	74.637

shut down. As a result, there is no continuity of power supply at the end of consumers. Adaptive protection coordination is required for all AC MG systems.

The results demonstrated that protecting an AC MG required a quicker execution time of 0.00202s. The proposed SVM algorithm approach program is a sufficient execution time and each DOC relay is properly coordinated with an AC microgrid. As a result, the execution time of the relay trip signals is important to the DOC relay on the faulty location system and trip signals are coordinated properly with the DOC relay on the AC MG system. The analysis of quicker program execution time to compare the proposed SVM algorithm approaches and all current machine-learning techniques. To optimize program execution time, the proposed SVM algorithm is employed. Because AC MG systems are highly protected, the SVM approach is suitable for real-time applications and large systems.

2) GRID-CONNECTED MODE OF OPERATION

The performance is evaluated using a grid-connected mode of operation. Different procedures are analyzed. (a) Fault Identification (FI) using the proposed method ANFIS (fault identification and different types of faults), (b) Comparison of error percentages (Faulty Location) between different ML techniques and the proposed approach, (c) Applying Different Machine Learning Approaches to Predict the Location of Faults in Grid-connected Mode, (d) Percentage of Error Estimation, and (e) Evaluation of Program Execution Time with Different Machine Learning Approaches.

a: COMPARISON OF ERROR PERCENTAGES ON AC MG GRID-CONNECTED MODE OF OPERATION USING VARIOUS ML APPROACHES

The ANFIS is very beneficial for analyzing complicated, nonlinear systems and making inference-driven judgments. Here's how to utilize ANFIS in grid-connected mode to find different types of faults in an IEEE12BUS AC MG system. First, membership functions for input variables such as voltage, current, etc. are defined. The linguistic variables "low," "medium," and "high" are represented by these membership functions, which characterize the states of each variable.

By collecting regular operations and different fault scenarios, membership functions are utilized to identify faults. Gradient descent and least squares approaches are combined

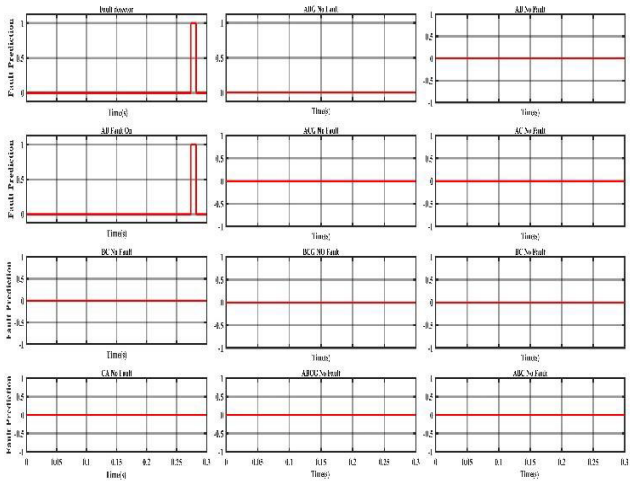

FIGURE 11. Fault identification is analyzed using the ANFIS technique.

in a hybrid learning strategy to optimize the ANFIS model parameters, including membership function parameters and rule weights.

The test system parameters are adaptively changed using training data, which is made up of input-output pairs (measurements of system variables and associated fault labels) to lower the error between expected and actual outputs.To classify the system's current state as normal faulty and identify the type of problem if present, the test system utilizes fuzzy logic inference based on the provided membership functions and methods.

The ANFIS test system's output enables prompt intervention and correction by providing information on the type of fault and likelihood of the problem found as shown in Figure 11. Through the use of generated or experimental data that represents a variety of fault scenarios, the effectiveness of the ANFIS-based fault identification system is assessed.

Adaptive Neuro-Fuzzy Inference Systems (ANFIS) must be able to adjust a significant variation in distributed generation (DG) and load in an AC microgrid. Here are some details on the potential constraints of ANFIS and how it can handle such variations:

- All possible practical distribution generations and various modes of operation (such as grid-connected and islanded) are taken into consideration while selecting the data set.
- The IEEE12BUS AC Microgrid model that has been developed is sufficiently resilient for operating in any kind of environment.
- The training dataset's quality and diversity have an enormous impact for effective ANFIS performance.
- When generating the dataset, researchers analysed the IEEE12BUS AC Microgrid system for every potential distribution generation, load conditions, and operating mode in many different situations.
- Nonlinear relationships between input variables (such as load demand and DG output) and control outputs

TABLE 5. Comparison of error percentages on AC MG grid-connected mode of operation using various ML approaches.

Data Points	Actual Faulty Location (Km)	SVM %Error	GRNN %Error	GPR %Error	NN %Error	EN %Error	BT %Error	DT %Error
1	5.7	-0.035	0	1.526	6.193	-3.544	1.93	0.105
2	15.84	-0.038	0	1.534	-5.568	1.376	1.787	0.732
3	25.52	-0.039	0	-1.532	-2.281	-14.001	-2.504	1.669
4	30.8	-0.039	1.201	-1.532	-2.955	-4.26	-4.003	0.101
5	8.8	-0.034	0	1.534	3.932	7.182	3.682	1.239
6	17	0.088	-0.353	1.535	-0.318	-9.529	-4.271	0.688
7	12	0	-4.25	1.533	1.2	14	2.85	0.1
8	22.05	0	-0.227	-1.533	-1.664	-4.93	-0.957	0.553
9	7.6	-0.039	0	1.539	6.197	0.842	3.684	0.105
10	14.08	-0.036	-5.54	1.534	1.229	13.999	0.838	2.23
11	21	0	-6.714	-1.533	-6.2	4.476	2.252	0.1
12	36	0.183	0	-1.533	-0.644	-0.111	-3.681	0.378
13	10.08	0	-0.7941	1.538	1.24	4.077	1.696	2.083
14	16	0.094	1	1.531	-1.2	7.75	3.681	0.1
15	24	0	0	-1.533	-2.033	-14	-2.013	1.35
16	31.68	-0.038	-1.61	-1.534	-5.568	-7.686	-3.681	0.417

(e.g. voltage and current) can be learned by utilizing ANFIS.

- By enhancing its membership functions and rules through training, it may adjust to changing situations.
- Variable renewable energy generation and dynamic load variations are two examples of the uncertainties and nonlinearities that ANFIS can manage.
- ANFIS can maintain system stability and improve performance in rapidly changing situations when trained properly.

b: COMPARISON OF ERROR PERCENTAGES ON AC MG GRID-CONNECTED MODE OF OPERATION USING VARIOUS ML APPROACHES

The estimated proposed SVM algorithm percentage error with different ML approaches for the AC MG islanded mode of operation is shown in Table 5. The lower and upper percentage error limits are 0.0 and 0.183 respectively. The SVM method faulty location is less than the compared GRNN, NN, Ensemble, GPR, BT, and DT for every positive percentage error value specified. The percentage of error negative value indicates that the SVM method fault distance is longer than the SVM technique.

The Bagged Trees (BT) technique is the lowest percentage of error analyzed with boundary lower and upper percentage error limits of 0.10 to 2.230, while the Ensemble approach is the highest percentage of error analyzed with boundary lower and upper percentage error limits of 0.111 to 14.00. The proposed SVM algorithm approach is the minimum and maximum percentages of error limitations of 0.0 and 0.183 respectively, when compared to all different ML approaches with the lowest percentage of error. The proposed SVM algorithm strategies are the most suitable approach. It is optimized for faulty locations using the SVM technique.

TABLE 6. Comparison of fault location on AC MG grid-connected mode of operation using various ML approaches.

Data Points	Grid-Connected Mode	Proposed Approach SVM Algorithm	ML Techniques Tools					
			GRNN	GPR	NN	Ensemble	BT	DT
1	Training Time (S)	**0.0022**	1.16845	310.5607	477.93	26.36092	283.11	17.4008

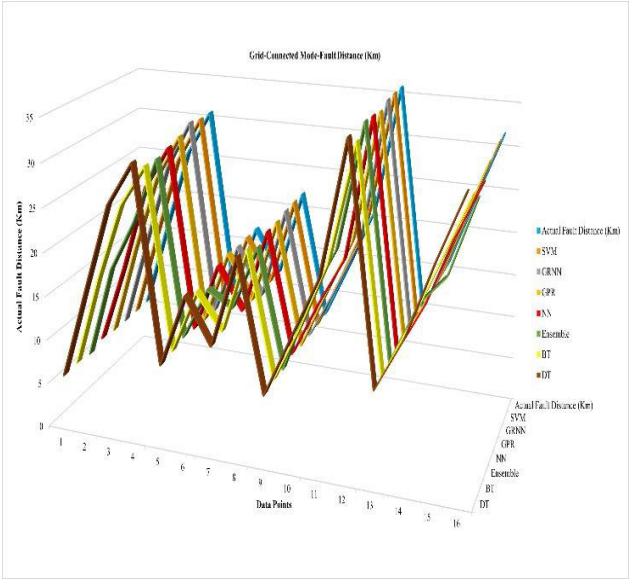

FIGURE 12. Comparison of different machine learning approaches to predict the location of faults in *Grid-connected Mode.*

c: APPLYING DIFFERENT MACHINE LEARNING APPROACHES TO PREDICT THE LOCATION OF FAULTS IN GRID-CONNECTED MODE

For better understanding, the comparison of several machine learning techniques for faulty location prediction in grid-connected mode and actual faulty location prediction is illustrated in Figure 12. The proposed SVM approach identified the faulty location for the AC MG system's Islanded mode with the highest percentage of accuracy in Table 6 and Figure 12. The ensemble is produced the worst results, while the predictions from GRNN, NN, GPR, ensemble, BT and DT are similarly closer. The ensemble method's observed error is very high, ranging up to 0.111% and 14.00%.

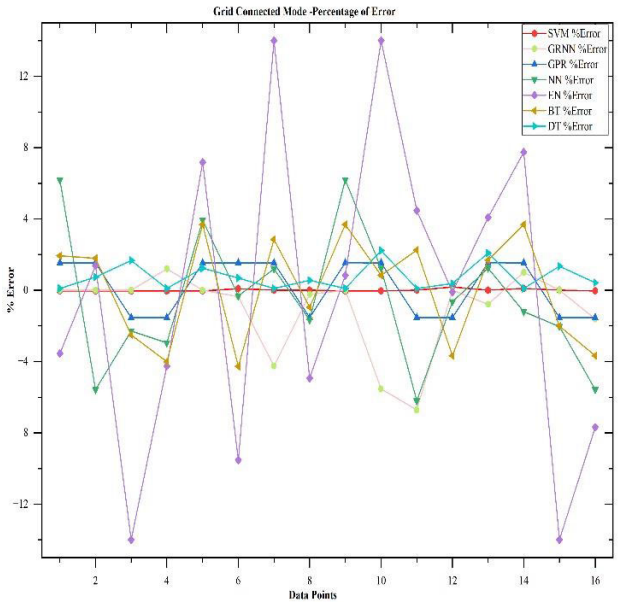

FIGURE 13. Estimation of percentage error graph grid-connected mode.

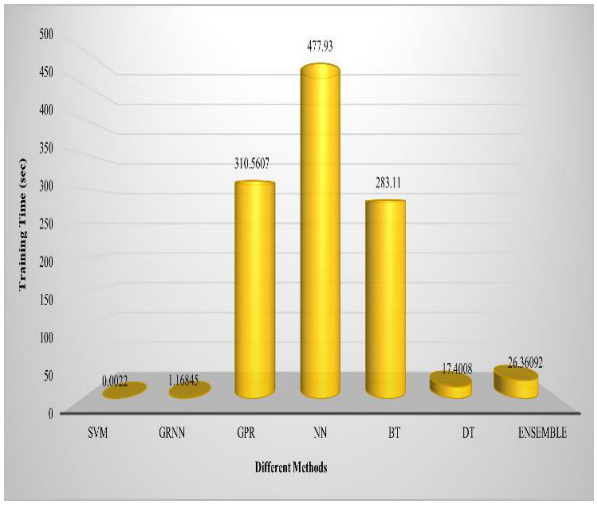

FIGURE 14. Evaluation of program execution time with different ML approaches grid-connected mode.

d: PERCENTAGE OF ERROR ESTIMATION

Figure 13 illustrates the suggested SVM algorithm and the expected percentage of errors for the AC MG grid-connected mode of operation applying several machine-learning approaches. The maximum and minimum percentage error limits are 0.167 and 0.021, the proposed SVM algorithm approach. The GRNN, GPR, NN, Ensemble, BT, and DT perform worse than the SVM technique for each positive percentage error number provided. The DT method's fault distance is greater than the SVM method, as indicated by the negative percentage of error value. The SVM approach is employed to optimize faulty location.

e: EVALUATION OF PROGRAM EXECUTION TIME WITH DIFFERENT MACHINE LEARNING APPROACHES

The proposed SVM algorithm technique is applied to estimate the program execution time in the faulty location on an AC Microgrid for a grid-connected mode of operation, as shown in Figure 14 and Table 7. The GRNN and NN methods produce the lowest percentage of errors but takes a longer program execution time (477.93s). The longer program execution time is causing the AC MG system to be out of synchronization. The entire AC MG system will shut down the power. As a result, there is no continuity of power supply at the end of consumers. Adaptive protection coordination is required for all AC MG systems.

TABLE 7. Comparison of training time on AC MG grid-connected mode of operation using different ML approaches.

Data Points	Actual Faulty Location (Km)	SVM	GRNN	GPR	NN	Ensemble	BT	DT
1	5.7	5.698	5.7	5.787	6.053	5.498	5.81	5.706
2	15.84	15.834	15.84	16.083	14.958	16.058	16.123	15.956
3	25.52	25.51	25.52	25.129	24.938	21.947	24.881	25.946
4	30.8	30.788	31.17	30.328	29.89	29.488	29.567	30.831
5	8.8	8.797	8.8	8.935	9.146	9.432	9.124	8.909
6	17	17.015	16.94	17.261	16.946	15.38	16.274	17.117
7	12	12	11.49	12.184	12.144	13.68	12.342	12.012
8	22.05	22.05	22	21.712	21.683	20.963	21.839	22.172
9	7.6	7.597	7.6	7.717	8.071	7.664	7.88	7.608
10	14.08	14.075	13.3	14.296	14.253	16.051	14.198	14.394
11	21	21	19.59	20.678	19.698	21.94	21.473	21.021
12	36	36.066	36	35.448	35.768	35.96	34.675	36.136
13	10.08	10.08	10	10.235	10.205	10.491	10.251	10.29
14	16	16.015	16.16	16.245	15.808	17.24	16.589	16.016
15	24	24	24	23.632	23.512	20.64	23.517	24.324
16	31.68	31.668	31.17	31.194	29.916	29.245	30.514	31.812

The results demonstrated that protecting an AC MG required a quicker execution time is 0.0022s. The proposed SVM algorithm approach program is a sufficient execution time, and each DOC relay is properly coordinated with an AC microgrid. As a result, the execution time of the relay trip signals is important to the DOC relay on the faulty location system and trip signals are coordinated properly with the DOC relay on the AC MG system. The analysis of quicker program execution time to compare the proposed SVM algorithm approaches and all current machine-learning techniques.

To optimize program execution time, the proposed SVM algorithm is employed. Because AC MG systems are highly protected, the SVM approach is suitable for real-time applications and large systems.

B. ANALYSIS OF PROPOSED APPROACHES FOR AN AC MG SYSTEM

The fault occurs in fault interaction between the fault lines and the AC MG system, the entire system will shut down.

As a result, there is no continuity of power supply at the end of consumers. Adaptive protection coordination is required for all AC MG systems. The performance of the presented research is tested on the IEEE12BUS AC microgrid system. The performance is analyzed for both grid-connected and islanded modes of operation. Execution times for islanded and grid-connected modes are 2.02 ms and 2.22 ms, respectively. The execution time of the proposed SVM algorithm approach program is sufficient, and each DOC relay is properly coordinated back-to-back with an AC microgrid. As a result of the significance of program execution time in communication with relay tripping time and fault signals. Due to the advantages of AC MG, it is used in many applications, such as control, monitoring, energy management, cyber security, and especially protection of AC MG for a reliable and safe operation.

C. COMPARATIVE ANALYSIS TO SUPPORT THIS APPROACH'S UTILIZATION OF ANFIS AND SVM

AC microgrids often exhibit nonlinear behavior due to the presence of various distributed energy resources and complex grid topologies. ANFIS is well-suited for modeling and understanding nonlinear relationships within such systems, enabling accurate fault identification even in challenging operating conditions. ANFIS can not only identify the presence of a fault but also estimate its identification within the grid. By analyzing spatial patterns in sensor measurements, ANFIS can pinpoint the affected area with high accuracy, facilitating targeted maintenance and restoration efforts.

ANFIS is robust to these uncertainties, thanks to its ability to handle fuzzy inputs and adapt its reasoning based on the available information, thereby enhancing the reliability of fault identification. ANFIS combines the advantages of fuzzy logic and neural networks, allowing it to capture both linguistic knowledge and numerical data. This hybrid approach enhances the system's ability to interpret vague and imprecise information, which is common in fault diagnosis scenarios.

ANFIS is a data-driven approach that learns from historical data to improve its fault identification capabilities over time. By training on a diverse dataset of fault scenarios, ANFIS can adapt its decision-making process to accurately distinguish between normal operating conditions and various fault types, including short circuits. Compared to traditional model-based approaches or expert systems, ANFIS offers a cost-effective solution for fault identification in AC microgrids. Once trained, ANFIS can operate autonomously with minimal human intervention, reducing the need for manual intervention and costly diagnostic equipment.

AC microgrids often exhibit complex and non-linear behaviours because of the integration of various distributed energy resources (DERs). Support vector machines are known for their capability to handle non-linear data and can be used to model and predict non-linear behaviour in microgrids. SVM is suitable to handle non-linear data, and as such, they can be utilized to model and anticipate non-linear behaviour in microgrids. SVM is efficient for handling high-dimensional data, can model complicated relationships among various variables, and can evaluate historical and real-time data to find patterns.

These features of SVM are appropriate for data-driven decision-making. SVMs can be employed for fault identification and relay operating time in the AC microgrids. They can aid in identifying the type and location of faults, allowing for a more precise and adaptive protection response. Certain types of faults might not occur often in microgrids, which could result in unbalanced datasets. By giving different

TABLE 8. Comparative analysis of state-of-the-art works.

SNo	Reference No	Models	Methods used	Evaluation metrics	
				Execution Time(S)	Percentage of Error
1	[1]	Abbasi, F., Abdoos, A.A., Hosseini, S.M., Sanaye-Pasand,2023	parallel lines if the parameter associated with the mutual coupling	Not computed	Maximum error= 0.5%.
2	[18]	Srivastava, A., Parida, S.K., 2022a	ML Learning Tools Technique, GPR, Ensemble, SVR, DT, BT	GPR= 5.5425, Ensemble=38.634, SVR=36.313, DT=4.2848	Minimal error = 0.064%, Maximum error = 0.86%
3	[19]	Srivastava, A., Parida, S.K., 2022b	Micro-PMUs and Phasor Measurement Devices (PMD).	GPR=38.2	Maximum error =0.763%
4	[15]	Rathore, B., Mahela, O.P., Khan, B., Alhelou, H.H., Siano, P., 2021	Proposed Algorithm ANN and STATCOM	Not computed	Minimal error = 0.87% Maximum error = 2.03%
5	[16]	Sapountzoglou, N., Lago, J., Raison, B., 2020.	Gradient Boosting Trees (GBT)	Not computed	Maximum error =0.3%
6	[8]	Gush, T., Bukhari, S.B.A., Mehmood, K.K., Admasie, S., Kim, J.-S., Kim, C.-H., 2019.	DOST-based optimized MKELM and KELM, SVM	Not computed	Minimal error = 01.17%, Maximum error = 3.87%
7	[12]	Majidi, M., Etezadi-Amoli, M., Sami Fadali, M., 2015	voltage magnitudes and phases measured by phasor measurement units (PMUs)	Not computed	Maximum error= 6.6%.
8	**Proposed model**	**Optimization of Fault Identification and Location**	**ANFIS and SVM at Islanded mode**	**SVM Algorithm Execution Time =2.02 ms**	**Minimal error = 0.00%, Maximum error = 0.167%**
			ANFIS with SVM at Grid-connected mode	**SVM Algorithm Execution Time =2.20 ms**	**Minimal error = 0.064% Maximum error = 0.183%**

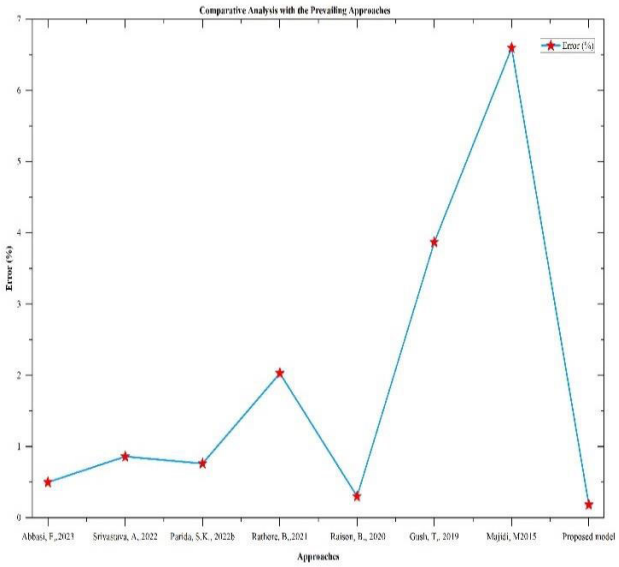

FIGURE 15. Comparative analysis with the prevailing approaches.

weights to classes, SVM can be modified to accommodate unbalanced datasets while making sure that frequent fault occurrences are not detected.

SVM can be used to calculate the most effective protection. In microgrids when the architecture of the system might shift frequently this is critical for adaptive protection coordination. SVM models can be utilized for real-time support for decision-making enabling adaptive coordination and quick response from the protection system to changes in the microgrid's operational conditions.

The AC MG system is efficiently coordinated, and the suggested methods provide outstanding results in terms of program execution time and DOC relay operation time. The percentage of error obtained from the proposed technique is 6.897 and is been compared with the recently published research publications. In this section, the performance of the presented research is analyzed with state-of-the-art works.

Table 8 and Figure 15 shows the comparative analysis of the proposed methodology with state-of-the-art works. In existing methodologies, the error value is lower for the proposed model, that is, 0.00 at islanded mode and 0.183 at grid-connected mode. The training time is much higher for the Ensemble system, that is, 38.634.

The state-of-the-art work based on the GPR, SVM, and DT ML Technique tools attains a 0.86% error. Here, the proposed approach provides better performance than the existing research methods. Based on the training time metric, the Fault Identification System and Faulty Location System-based existing research is 38.634s at the faulty location. GPR-based existing work is 38.2s training time. The proposed SVM algorithm approach Low Computational Burden Fault Identification System and Faulty Location System is 2.02 ms and 2.20 ms in both islanded and grid-connected modes. However, the proposed hybridized model attains better performance in both islanded and grid-connected modes.

D. LIMITATIONS

Determining AC microgrid constraints is essential for testing the proposed research project and efficiently identifying and locating faults. The performance and viability of AC microgrids can be improved by researchers concentrating on resolving three important constraints. Keeping distributed energy resources (DERs) in synchronicity concerning voltage, current, and phase. It is still difficult and computationally demanding to coordinate several DERs and loads in a decentralized system. Sophisticated synchronization and stability control methods are as follows.

- Protection plans that are high impedance faults for systems with results in low fault current.
- Improved protection coordination with renewable energy storage devices.
- Development of economical, scalable AC microgrid systems.
- Creation of regulatory structures that are supportive and universal norms.
- The proposed approach work is well-suited for AC Microgrid systems, which operate in real-time, especially in bigger systems such as the IEEE 14-Bus and IEEE 35-Bus.

IV. CONCLUSION

This work utilizes machine learning techniques to detect and locate faults within an AC Microgrid (MG) system that contains Distributed Generators (DGs). It assumes access to one-cycle faulty voltage and current measurements at the source buses via Smart Digital Relays (SDRs) systems. The proposed methodology utilizes these measurements to develop a Fault Identification System (FIS) using Adaptive Neuro-Fuzzy Inference System (ANFIS) techniques.

In testing the approach, the IEEE12BIS AC MG model demonstrates highly accurate fault identification, pinpointing both the faulted line type and its fault location with precision. Moreover, a comparative analysis of accuracy validates the superior performance of the ANFIS system when contrasted with traditional methods such as Bagged Trees (BT) and Decision Trees (DT).

The same inputs are then employed to develop the FLS using the SVM algorithm approach, which resulted in the most accurate faulty location, with percentages of error minimum and maximum limits **of 0.021% and 0.167%** for islanded mode and **0.00% and 0.183%** for grid-connected mode, respectively.

The SVM method provided accurate results as compared to ML method tools (GRNN, GPR, NN, Ensemble, BT, and DT) and commonly used faulty location techniques viz. The proposed SVM algorithm is a minimum RMS error of 0.003 with the least execution times of **2.02ms** for islanded mode and **2.20ms** for grid-connected mode, which tested that it is a genuine tool for faulty location estimation.

The AC MG system is well-coordinated, and the proposed approach produces excellent results in terms of program execution time for both island and grid-connected modes.

The aforementioned findings and discussions indicate that the SVM algorithm technique is suitable for real-time applications of an AC MG system. However, the proposed research analyzed different faulty locations to analyze different faults for the IEEE12BUS AC MG system.

NOMENCLATURE

Ns	Distribution Networks.
AC	Alternating Current.
DG	Distribution Generation.
DOCRs	Directional Over-Current Relays.
ML	Machine Learning.
GPR	Gaussian Process Regression technique.
HHT	Huang Hilbert Transform.
DWT	Discrete Wavelet Transform.
DFT	Discrete Fourier Transform.
DOST	Discrete Orthonormal Stock well Transform.
DSP	Digital Signal Process.
ANNs	Artificial Neural Networks.
KNN	K-Nearest Neighbor.
DT	Decision Trees.
RMG	Renewable Microgrids.
IPS	Intelligent Protection System.
BT	Bagged Trees.
KV	Kilo Volt.
MVA	Mega Volt-Ampere.
ω_s	Relay time.
t_{if}	Relay operating time.
Q^n	Vector space.
$R_{p.u}$	Resistance unit value.
I_{ft}	Fault current.
I_{bt}	Base current.
ANFIS	Neuro-Fuzzy Inference System.
SVM	Support Vector Machine.
FIS	Fault Isolation System.
FLS	Fault Location System.
PCC	Point of Common Coupling.
RMS	Root Mean Quire.
GRNN	Generalized Regression Neural Network.
NN	Neural Network.
MG	Microgrid.
$\Delta V_c(g)$	Mean deviation in phase voltage.
$\Delta I_c(g)$	Mean deviation in frequency.
$C^-_{v,x}(g)$	Undervoltage counter.
$C^+_{v,x}(g)$	Overvoltage counter.
$T^-_{v,x}$	Undervoltage threshold.
$T^+_{v,x}$	Overvoltage threshold.
Fr	Frame rate.
T_{lag}	Time delay between RMS and RDC processing.
T_{fault}	Fault-tolerant time.
$\frac{C^-_{v,x}(g)}{F_r}$	Disturbance counter.

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The authors declare that not applicable.

AUTHOR CONTRIBUTIONS

A. Kurmaiah: Conceptualization, design and development, methodology, software, data curation, examined, writing-original draft preparation, visualization, investigation. C. Vaithilingam: Supervision, software, validation, resources, writing-reviewing and editing, data curation. All authors read and approved the final manuscript.

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